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Effects of New Zealand blackcurrant, body composition, and sex on fat oxidation during exercise

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Abstract

New Zealand Blackcurrant (NZBC) has a high anthocyanin content and intake has been shown to increase fat oxidation during 30-min exercise, with magnitude of change correlated to body composition in males and females. No investigation has compared males and females fat oxidation responses to NZBC intake during longer duration exercise within the same study. PURPOSE: To examine the effect of body composition and NZBC extract on metabolic responses during 60-minutes treadmill running in males and females. METHODS: Twenty-two (11 male and females, mean $\pm$ SD, age: 29 $\pm$ 8 years, height: 171 $\pm$ 10 cm, mass: 69 $\pm$ 11 kg, body fat: 18 $\pm$ 6 %, $\dot{V}O_{2max}$: 44 $\pm$ 9 mL $\cdot$ kg $^{-1}\cdot$ min $^{-1}$) participants underwent a double-blind, randomised cross-over and placebo-controlled study. Participants consumed NZBC extract (600 mg $\cdot$ day $^{-1}$ containing 210 mg anthocyanin) or placebo (PLA) for 7-days (washout 14-days). Participants performed 60-minutes of treadmill running at 50% $\dot{V}O_{2max}$ with measurement of expired gases (Cortex Metalyzer 3B, Leipzig, Germany). Bioelectrical impedance measured body composition (Bodystat 1500, Douglas Isle of Man). RESULTS: There was no difference in relative intensity for NZBC and placebo for males (NZBC: 51.7 $\pm$ 6.1, PLA: 49.5 $\pm$ 6.9) and females (NZBC: 52.1 $\pm$ 6.9, PLA: 52.9 $\pm$ 9.2 % $\dot{V}O_{2max}$) ($P=0.109$). Average fat oxidation was different between the conditions (NZBC: 0.27 $\pm$ 0.11, PLA: 0.21 $\pm$ 0.12 g $\cdot$ min $^{-1}$, $P<0.001$), but the responses between males (NZBC: 0.27 $\pm$ 0.11, PLA: 0.19 $\pm$ 0.11 g $\cdot$ min $^{-1}$) and females was not different (NZBC: 0.26 $\pm$ 0.12, PLA:

0.22±0.14 g·min⁻¹, P=0.078). When combined, there was no relationship (P>0.05) between body fat percentage, fat-free-mass, and fat-mass for change in fat oxidation (Δ FATOX) (r=-0.079, r=0.141, r=0.012, respectively). For males there was no relationship (P>0.05) between body fat percentage, fat-free-mass, and fat-mass for Δ FATOX (r=-0.069, r=0.039, r=0.029, respectively). However, for females there was a relationship between body fat percentage and Δ FATOX (r=0.691, p=0.019) but no relationship (P>0.05) between fat-free-mass and fat-mass for Δ FATOX (r=-0.429, r=0.538, respectively). CONCLUSION: A 7-day intake of New Zealand blackcurrant extract increases fat oxidation during 60-min moderate-intensity exercise, with the change related to body composition in females.

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